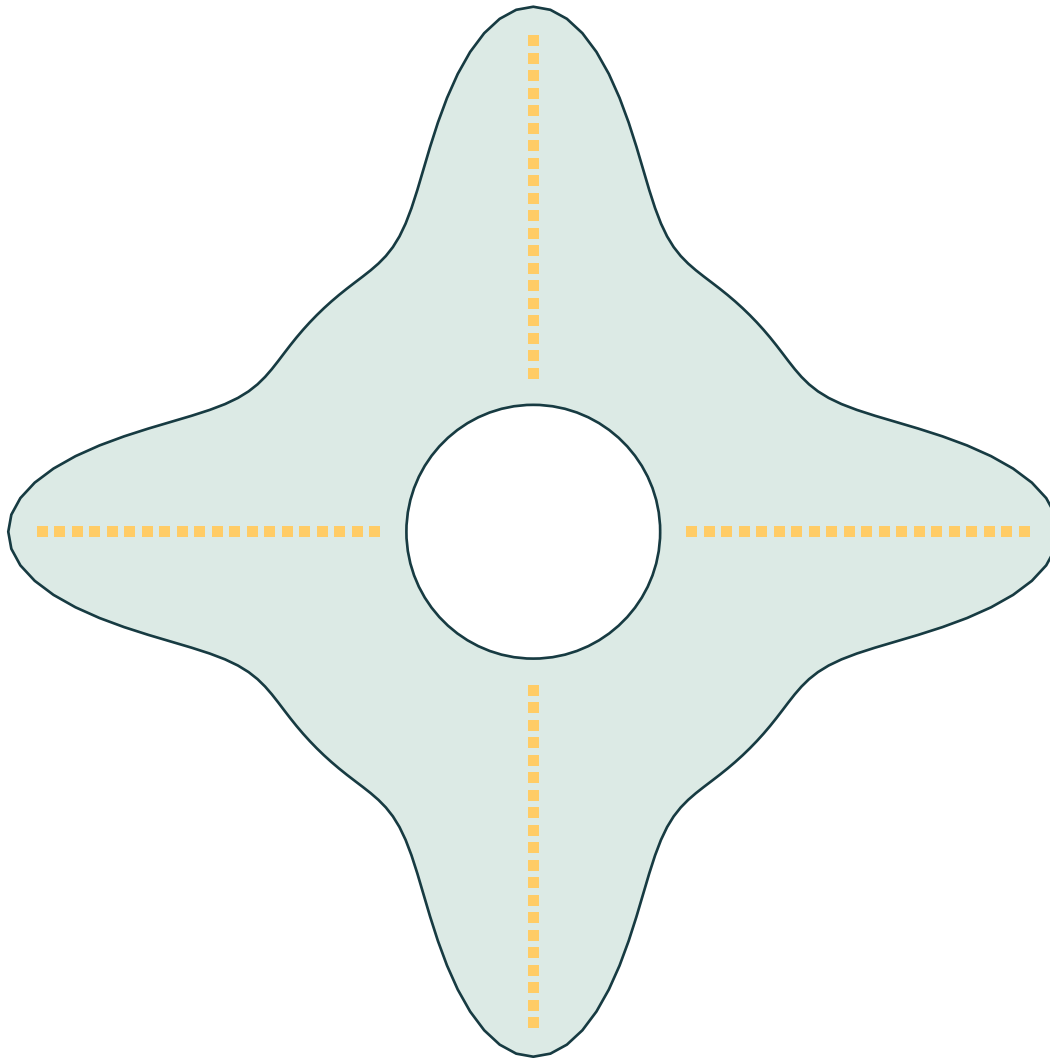


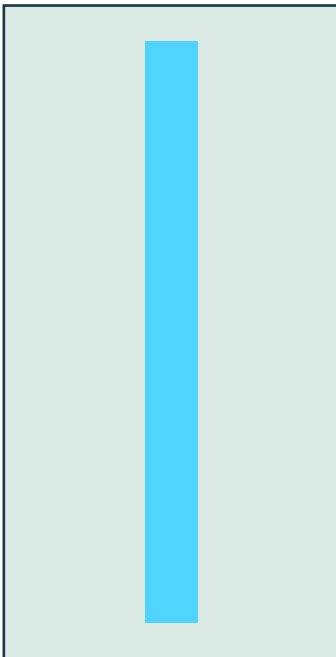


MAIN PCB NOTES

4 layers, finished thickness 1.00 +/- 0.10 mm
Overall envelope 96.0 mm; center hole 23.20 mm
Four M2 NPTH holes: 2.40 mm at radius 17.0 mm / 45 deg
ENIG recommended; lead-free assembly
Minimum track/space 0.15/0.15 mm; smallest top LED pads 0.48 mm
RF keep-out: no copper, vias or metal enclosure above module antenna

STACKUP / ACCEPTANCE

L1 signal + LED distribution
L2 solid GND reference
L3 +5V_LED / +3V3 power islands
L4 signal + components
Supplier may propose equivalent controlled stackup
100% flying probe, AOI after SMT, first-article dimensional report



TIP BOARD NOTES

Quantity: four identical boards per spinner
2 layers, finished thickness 0.60 +/- 0.07 mm
Finished size 7.00 x 13.60 mm
12 x SK6805-EC10, 1.00 mm pitch
Panelize with rails; routing tabs outside light window
ENIG required; confirm LED orientation with first article

MECHANICAL INTERFACE

Board slides into enclosure guide; enclosure carries hand impact
Four solder tabs mate to main board: +5V, GND, DIN, DOUT
Maintain 0.20 mm board-edge clearance from copper
Inspect every tip board with walking-one RGB fixture
Do not substitute a taller LED without checking 16 mm enclosure stack